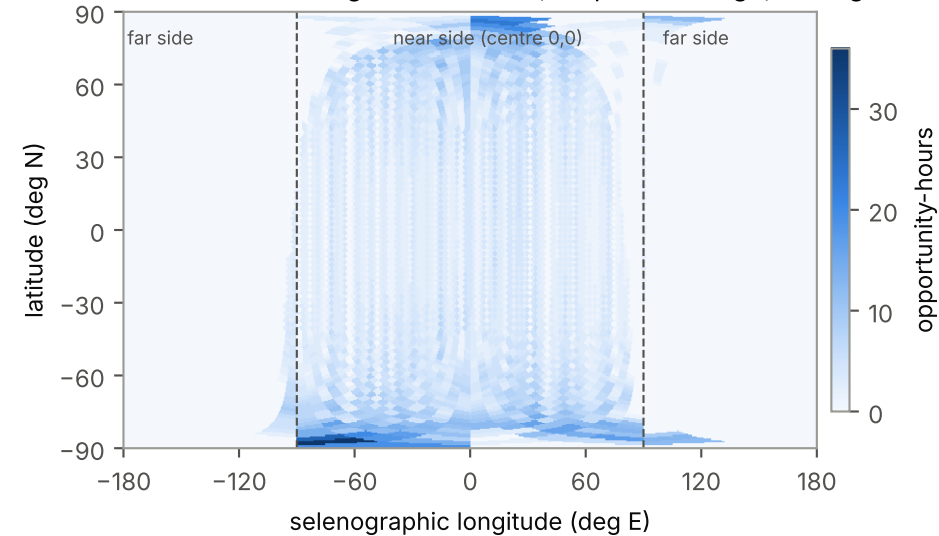
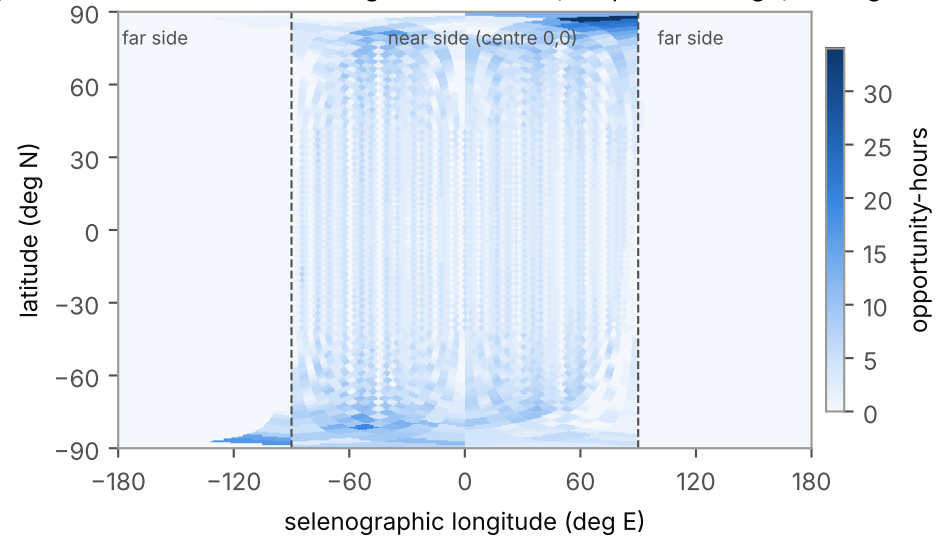


Conditional reachability: hours in the impact domain when a pixel is on the ground track (polar 100-km orbit), dark, Earth-facing, phase 3-65 %, and observable

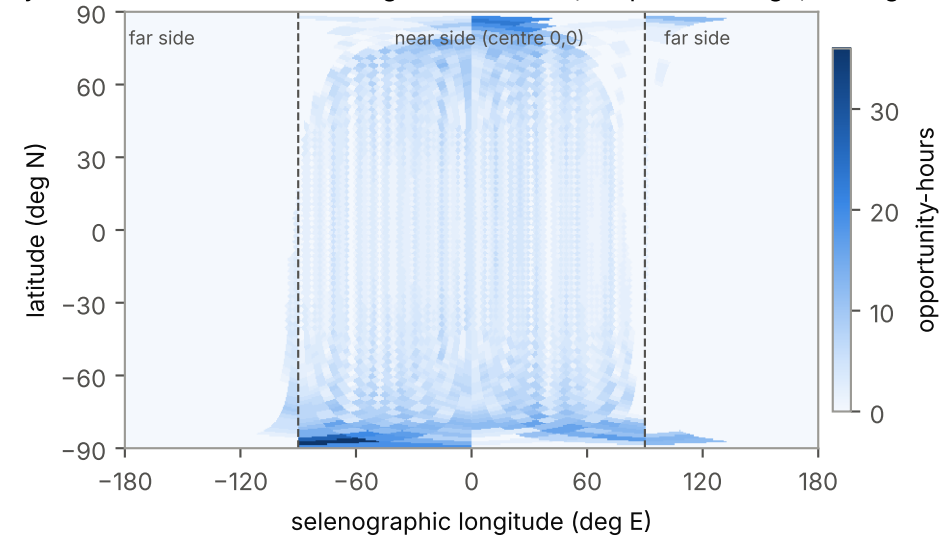
a) node lon. 0 deg (2027-09-01), no plane change; flash geometry + Turkish site



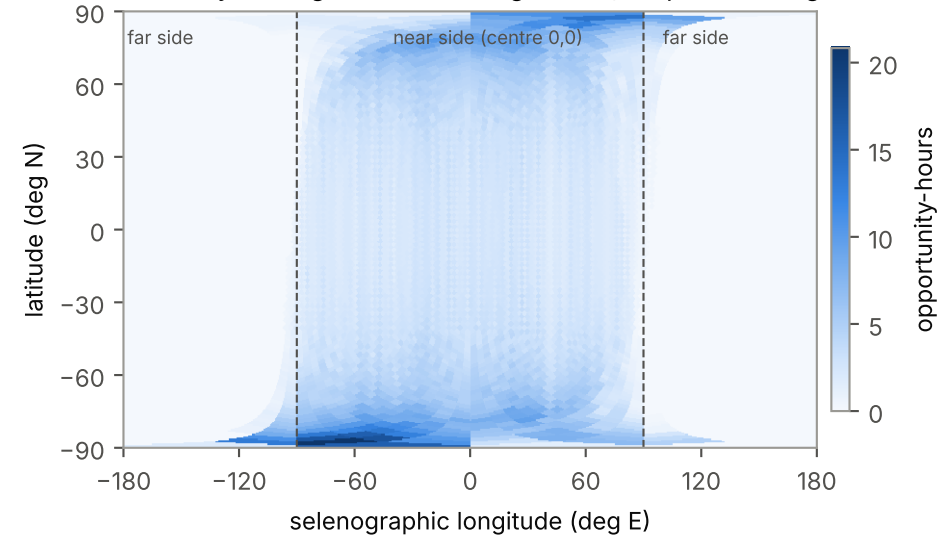
2) node lon. 90 deg (2027-09-01), no plane change; flash geometry + Turkish site



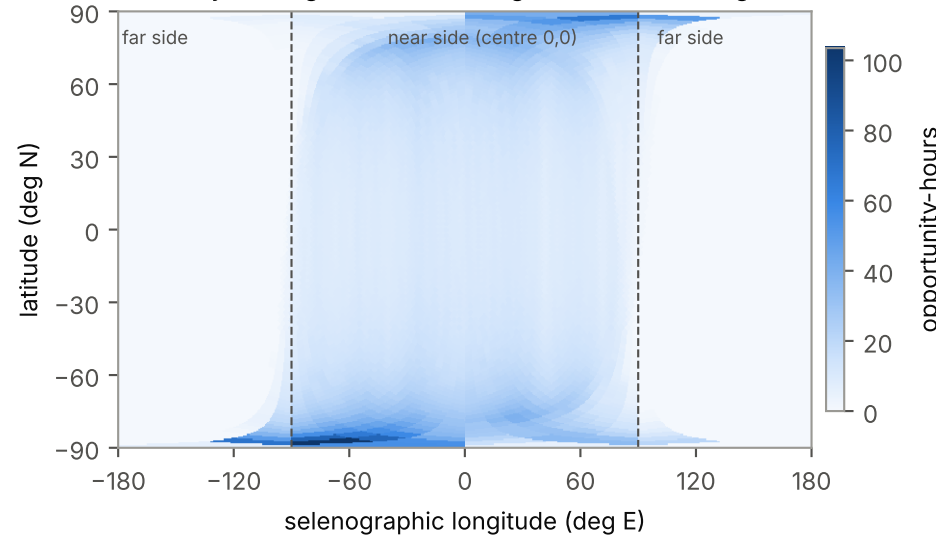
3) node lon. 180 deg (2027-09-01), no plane change; flash geometry + Turkish site



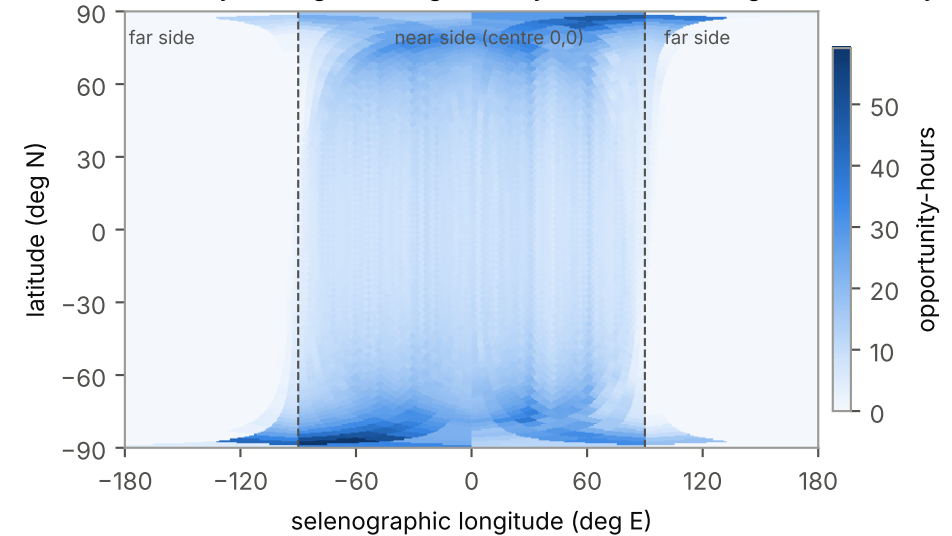
b) family average (24 node longitudes), no plane change



c) family average with a 2.5-deg cross-track budget (~70 m/s)



d) family average: flash geometry with ≥ 3 configured sites anywhere



CONDITIONAL on declared orbit-plane families - the real AYAP-1 orbit plane is unpublished